

LESSON 3

INIT, Telemetry, and waitForStart()

Follow execution through Driver Station states

MISSION Can you identify the exact line where the program waits before PLAY?

Learning targets

CHECK	BY THE END OF THIS LESSON, I CAN...
[]	Describe what executes during INIT.
[]	Compare telemetry.addData() and update().
[]	Explain waitForStart().
[]	Describe how PLAY and STOP change execution.

Words to know

VOCABULARY runOpMode | INIT | telemetry packet | update | waitForStart | PLAY | active | STOP

Code focus

```
telemetry.addData("Status", "Initialized");
telemetry.update();

waitForStart();

if (opModeIsActive()) { ... }
```

Before the video: commit to a prediction

MY PREDICTION Before PLAY, will moving the joystick execute the setPower() lines? Explain why or why not.

Confidence before the lesson: 1 2 3 4 5 (circle one)

The POWERING ON safety contract

NON-NEGOTIABLE The sequence is completed before every live-motor test. If a step is skipped, remain disabled and restart at Step 1.

STEP	ACTION	TEAM CHECK
1	SECURE THE ROBOT Use stable blocks. Keep driven wheels and moving mechanisms clear of the table and floor.	[]
2	CONFIRM DISABLED Select the correct robot and OpMode, but do not press PLAY.	[]
3	LOOK AROUND Remove hands, tools, wires, scraps, and loose objects from moving areas.	[]
4	MAKE SURE ALL ARE CLEAR Everyone steps outside the boundary; secure hair, clothing, lanyards, strings, and jewelry.	[]
5	CALL AND WAIT Call POWERING ON! loudly. Pause long enough for everyone to stop, look up, and step away.	[]
6	VISUALLY CONFIRM, THEN ENABLE Scan the full area again. Press PLAY only when everyone and everything is clear.	[]

Callout script

BEFORE PLAY Robot is secured and DISABLED. Look around. Everyone clear? POWERING ON! ... Wait ... Area is visibly clear. Enabling now.

BEFORE CONTACT STOP pressed. Driver Station shows DISABLED. Hands may approach only after the operator confirms disabled.

Student agreement

I understand that anyone may call STOP and the operator will disable immediately. I will not approach or touch the robot until DISABLED is visually confirmed.

Student signature: _____ Date: _____

Reference program: BasicTwoMotorTeleOp.java

Use this numbered reference throughout Lessons 1-6. The comments divide the program into startup and repeating control sections.

```

01 package org.firstinspires.ftc.teamcode;
02
03 import com.qualcomm.robotcore.eventloop.opmode.LinearOpMode;
04 import com.qualcomm.robotcore.eventloop.opmode.TeleOp;
05 import com.qualcomm.robotcore.hardware.DcMotor;
06
07 @TeleOp(name="Basic Two Motor TeleOp", group="Iterative Opmode")
08 public class BasicTwoMotorTeleOp extends LinearOpMode {
09     private DcMotor leftMotor;
10     private DcMotor rightMotor;
11
12     @Override
13     public void runOpMode() {
14         leftMotor = hardwareMap.get(DcMotor.class, "left_motor");
15         rightMotor = hardwareMap.get(DcMotor.class, "right_motor");
16
17         // Set one motor REVERSE if the physical drivetrain requires it.
18         // rightMotor.setDirection(DcMotor.Direction.REVERSE);
19
20         telemetry.addData("Status", "Initialized");
21         telemetry.update();
22         waitForStart();
23
24         if (opModeIsActive()) {
25             while (opModeIsActive()) {
26                 double leftPower = -gamepad1.left_stick_y;
27                 double rightPower = -gamepad1.right_stick_y;
28                 leftMotor.setPower(leftPower);
29                 rightMotor.setPower(rightPower);
30                 telemetry.addData("Left Power", leftPower);
31                 telemetry.addData("Right Power", rightPower);
32                 telemetry.update();
33             }
34         }
35     }
36 }

```

CODE DETECTIVE Circle `waitForStart()`. Box the while condition. Underline both `setPower()` calls. Draw a star beside each telemetry line.

LESSON

3

Guided Video Notes

INIT, Telemetry, and waitForStart()

DIRECTIONS At each timestamp, pause, think silently, discuss, and then record evidence from the code or whiteboard visual.

PAUSE	QUESTION	MY CODE / VISUAL EVIDENCE
0:17	What causes runOpMode() to begin?	
0:27	What setup happens before waitForStart()?	
0:46	How do addData() and update() differ?	
0:55	Does Initialized mean the driving loop is running?	
1:05	What does waitForStart() wait for?	
1:34	What changes after PLAY?	
1:53	What happens to opModelsActive() after STOP?	

MOST IMPORTANT IDEA The idea I would teach to a teammate is:

LESSON

3

Hands-On Team Activity

INIT, Telemetry, and waitForStart()

Materials and setup

READY	ITEM
[]	Robot secured on blocks; gamepad and Driver Station connected.
[]	BasicTwoMotorTeleOp.java displayed with telemetry and waitForStart() visible.
[]	Printed state-observation sheet from this guide.
[]	Cards labeled BEFORE INIT, INITIALIZED, WAITING, ACTIVE, and STOPPED.
[]	POWERING ON safety poster displayed beside the robot area.

Team roles

ROLE	JOB
Safety lead	Controls the call-and-clear sequence.
Driver Station	Presses INIT, PLAY, and STOP when authorized.
Code tracker	Points to the instruction where execution is located.
Telemetry reader	Reads each displayed telemetry message aloud.
Wait gate	Blocks progress until PLAY is pressed.
Joystick tester	Moves one stick only when the teacher authorizes it.
Recorder	Completes the state-observation table.
Stop verifier	Confirms the OpMode ended and the robot is DISABLED.

Activity sequence

STEP	ACTION
1	Place the four state cards in order. Ask students to predict whether motor commands can repeat in each state.
2	With the robot disabled and secured, select the OpMode and press INIT. Ask the code tracker to follow runOpMode() until execution pauses.
3	Ask the joystick tester to move a stick before PLAY while everyone watches from outside the boundary.
4	Run the complete POWERING ON sequence. After the pause and visual clear check, authorize PLAY and one gentle stick movement.
5	Direct the Driver Station operator to press STOP. Ask the stop verifier for two pieces of evidence.

LESSON 3

Lab Record and Exit Ticket

INIT, Telemetry, and waitForStart()

Student/team: _____ Date: _____ Robot: _____

BEFORE PLAY	AFTER TEST
☐ Robot secure ☐ DISABLED ☐ Area clear ☐ POWERING ON called ☐ Wait completed ☐ Visual clear	☐ STOP pressed ☐ DISABLED visually confirmed Safety lead initials: _____

Program-state observation record

STATE	CODE LOCATION	MOTOR LOOP?	OBSERVED EVIDENCE
Before INIT	Not running	No	
After INIT	Setup -> waitForStart()	No	
Stick moved before PLAY	Still waiting	No	
After PLAY	Active check / while loop	Yes	
Stick returned to center	Loop continues	Yes; power may be 0	
After STOP	Loop exited	No	
DISABLED confirmed	OpMode ended	No	Initials: _____

Debrief

#	QUESTION	MY RESPONSE
1	Why can Initialized be displayed before the robot can drive?	
2	What is the difference between centered sticks and STOP?	
3	What does telemetry.update() do?	
4	What proves it is safe to approach?	

Exit ticket

ANSWER BEFORE LEAVING Why can Status: Initialized be true while joystick movement still produces no wheel response?
